STA 325 Case Study Report

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Introduction

This case study is an exploration of how 3 factors affect the clustering of inertial particles in a turbulent flow such as clouds. The statistical modeling uses a dataset produced by computer simulation. The dataset has 89 observations of clustering behaviors, where each response was observed under the following variables:

- **Reynolds number** Re: 90, 224, 398.
- **Froude number** Fr: 0.052, 0.3, ∞.
- **Stokes number** St: 14 values between 0.05 and 3.

Our key research objectives are:

1. Predicting the mean, standard deviation, skewness, and kurtosis of the distribution at a new (Re, Fr, St) observation
2. Explaining how each parameter affects the response

Our final model is as follows, where Re_{224}, Re_{398}, $Fr_{0.3}$, and Fr_{∞} are 0/1 indicators for each level (baseline $Re = 90$, $Fr = 0.052$).

$$\begin{aligned}\log \text{responses} = \beta_0 + \beta_1 \log St + \beta_2 Re_{224} + \beta_3 Re_{398} + \beta_4 Fr_{0.3} + \beta_5 Fr_{\infty} \\ + \beta_6 Re_{224} Fr_{0.3} + \beta_7 Re_{224} Fr_{\infty} + \beta_8 Re_{398} Fr_{\infty} + \varepsilon, \qquad \varepsilon \sim N(0, \sigma^2)\end{aligned}$$

This model addresses both objectives. For prediction, it achieves a LOOCV MSE of 0.122 when averaged across the 4 response variables. For inference, it is a linear model, so each coefficient is easy to interpret and understand.

Methodology

Responses and predictors

Each observation in the dataset reports the first four raw moments of the response distribution, which we converted to central moments with `raw2central`. From these we computed the mean, standard deviation, skewness, and kurtosis of each distribution, which are the 4 desired response variables.

Modeling

We selected our final model using two steps.

We first compared linear modeling with generalized additive modeling (GAM). The resulting MSE values showed comparable performance levels. We decided to use linear modeling as linear models are more easily interpreted and generally have lower variance compared to GAMs, which is valuable on unseen data.

We fitted 3 linear models for comparison. All 3 models predict the log of the response variables. The baseline model has no interaction terms. Interaction Model 1 adds interactions of $\log St$ with both Re and Fr, as well as $Re \times Fr$. Interaction Model 2 only adds $Re \times Fr$.

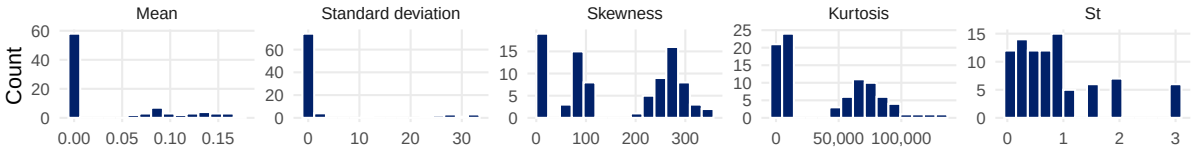
$$\begin{aligned}
\text{Baseline:} \quad & \log y = \log St + Re + Fr \\
\text{Interaction Model 1:} \quad & \log y = \log St + Re + Fr + \log St \times Re + \log St \times Fr + Re \times Fr \\
\text{Interaction Model 2:} \quad & \log y = \log St + Re + Fr + Re \times Fr
\end{aligned}$$

Averaged across the 4 response variables, the LOOCV MSE was 0.609 for the baseline model, 0.156 for Interaction Model 1, and 0.122 for Interaction Model 2. Interaction Model 1 had a slightly lower LOOCV MSE for the mean, but Interaction Model 2 performed better for the other 3 response variables, so we selected Interaction Model 2 as our final model.

Model	Parameters	Mean	Standard deviation	Skewness	Kurtosis
Baseline (additive)	6	0.0192	0.7338	0.3561	1.3252
Interaction Model 1	13	**0.0104**	0.3290	0.0715	0.2119
Interaction Model 2	9	0.0114	**0.2565**	**0.0554**	**0.1656**

For each of the models four separate linear fits are created, one for each of the response variables. LOOCV fits each of those models 89 times, leaving out one observation for each fit.

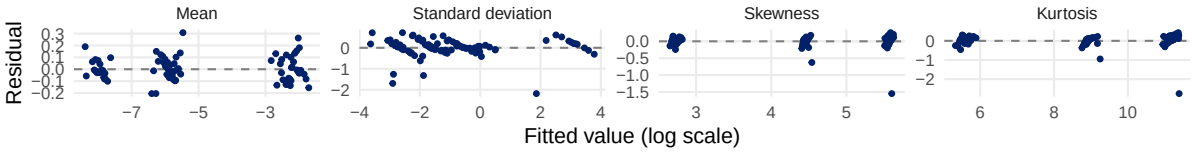
We decided to log-transform the response variables because they are right-skewed and span wide ranges. We also log-transformed St for the same reason.



Re and Fr each have 3 levels in the dataset, so we treat them as categorical.

Diagnostics

The final model fits all 4 response variables well, with R^2 between 0.942 and 0.998.



The residuals are centered around 0 with roughly equal vertical spread and no significant pattern, so the linearity and equal variance assumptions are satisfied. The dataset was collected from independent computer simulation runs, so the independence condition is also satisfied. $n = 89 > 30$ also satisfies the normality assumption.